AKARSHA JAIN
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EDUCATION

University of Southern California , *Los Angeles, USA* — Masters in Computer Science — **Jan, 2025 – Dec, 2026**
Walchand Institute of Technology, *Solapur, India* — Bachelor of Technology In Computer Science — **Aug, 2019 – Jun, 2023**

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, Kotlin, C, R, SQL, TypeScript, Bash
Machine Learning & AI: TensorFlow, PyTorch, YOLOv8, OpenCV, Scikit-learn, NumPy, Pandas, Matplotlib, BERT, NLP, CNNs, GPT APIs, Hugging Face Transformers, XGBoost, Optuna
LLM & NLP Ecosystem: Llama 3, Mistral-7B, Falcon, Ollama, SentenceTransformers, LangChain, Prompt Engineering, Tokenization, Quantization, RAG Pipelines, FAISS, ChromaDB
MLOps & Data Engineering: Docker, Kubernetes, Helm, Apache Kafka, Multiprocessing, Model Quantization, Data Pipelines, MLflow, OpenVINO, ONNX, Airflow (familiar), Spark (familiar)
Cloud Platforms: AWS (EKS, ECS, EC2, S3, RDS, EFS, SageMaker, Lambda), Serverless, GCP (familiar)
Data & Analytics: Excel, Tableau, Google Sheets, Data Visualization, Plotly, R Shiny, Seaborn
Software Engineering: Spring Boot, Flask, FastAPI, Node.js, React.js, REST APIs, Microservices, Agile SDLC, Design Patterns
Databases: PostgreSQL, MySQL, SQL Server, Redis, Flyway (Schema Migration)
Testing & Security: JUnit, Mockito, SonarQube, OAuth2, JWT, Espresso | **Version Control & CI/CD:** Git, GitHub Actions, GitLab, Jenkins, Harness
Research Tools: Survey Design, Qualitative & Quantitative Coding, Benchmarking, Reporting
Tools & OS: Jupyter, VS Code, IntelliJ, Linux/Unix, JIRA

WORK EXPERIENCE

USC Summer Program (USC Viterbi K–12 Program)| *Teaching Assistant* | *Los Angeles, California, United States* — **May 2025-Present**

- Taught and mentored **60+ students** in **AI-driven robotics programming** using **Zumi AI, Red Bot, EZ-Bots, and Arduino**-based platforms for autonomous navigation and disaster recovery simulations.
- Guided integration of **computer vision, sensor fusion, and ML-based object detection** with ultrasonic, bumper, line-following, and thermal sensors for real-time obstacle avoidance and victim identification.
- Conducted **Python programming** sessions for **thermal-sensor drones**, integrating **IoT sensor data** from AirBeam3 and Databot kits to simulate **environmental monitoring** and **data-driven decision making**.
- Advised teams on **data-driven robotics design**, IEEE-format research paper writing, and web portfolio development showcasing code, design process, and AI/ML integration.

National Australia Bank (NAB Innovation Center India) | *Software Engineer* | *Gurugram, Haryana, India* — **July 2023-December 2024**

- Engineered and maintained **6+ Java/Spring Boot microservices** processing **1M+ daily transactions** with **<100ms latency**, **ensuring real-time fraud detection** responsiveness.
- Reduced transaction latency **by 40%** through **Redis caching, JDBC** connection pool tuning, and SQL optimization on **5M+ records**.
- Designed real-time alerting pipelines **with Apache Kafka** Streams to **improve fraud detection accuracy by 25%** and **developed secure 3DS authentication** APIs using OAuth2 and JWT for **multi-wallet fraud reconciliation**.
- Containerized microservices with **Docker** and automated zero-downtime **Helm** deployments on **AWS EKS**, sustaining **99.99% uptime**; resolved **JVM warm-up** latency issues, earning **Spot Award**.

Persistent Systems | *Martian Program PSPL Intern* | *Pune, Maharashtra, India* — **May 2022-June 2022**

- Developed a single-page **React.js** application with dynamic routing powered by React Router, ensuring smooth navigation and fast page loads.
- Built **RESTful APIs** using **Node.js** and **Express.js** backend, implementing **JWT-based authentication** with secure token storage and refresh logic.
- Implemented persistent authentication state across sessions using localStorage and context API, preventing user logout on page reloads.
- Conducted **cross-browser testing** and **debugging** to ensure consistent user experience across Chrome, Firefox, and Safari.

PROJECTS

Smart Glasses | Incubated under TATA Technologies
Tools: *TensorFlow, OpenCV, YOLOv8, CNN, EdgeTPU, Raspberry Pi, Google TTS, GPS*

- Engineered smart glasses for visually impaired users using Raspberry Pi 5, camera module, **YOLOv8**, and **CNNs** for real-time object detection, with precise classification via **OpenCV** and **TensorFlow**. ***Winner, TATA MAGIC Innovation Challenge.***
- **Integrated NLP** and **Google TTS** to convert detected visual objects into spoken feedback through in-ear audio; optimized for low power consumption and scalability with **facial recognition and GPS navigation**.

Toy car Automation using ML
Tools: *Raspberry Pi, YOLOv8, OpenCV, TensorFlow, GPIO Sensors, Python*

- Developed an autonomous toy car powered by **Raspberry Pi**, **YOLOv8**, camera, and multiple sensors for **real-time object detection**, **obstacle avoidance**, and road navigation.
- Implemented **computer vision algorithms** and **GPIO motor control** to enable intelligent path planning, demonstrating expertise in **ML, computer vision (CV), and embedded systems**.

LEADERSHIP & COMMUNITY ENGAGEMENT

- Graduate Student Affairs Officer, **Society of Women Engineers (SWE) @ USC |** Founder & Lead, **Coding Club**, Walchand Institute of Technology.
- Awarded funding and won in the Magic competition with team Electra, in collaboration with **TATA Technologies Ltd**.
- **Received Spot Award** recognition on a quarterly basis for exceptional team performance and critical production issue resolution.